

Introduction to Tools & Techniques

100 Questions: ML Fundamentals, Data Types, Pre-processing, PCA

GARP RAI Exam Preparation

Q1: Machine Learning Definition - Tutorial

QUESTION: What is Machine Learning (ML)?

OPTIONS:

- ❶ A method for storing large datasets
- ❷ An umbrella term for techniques that train models to recognize data patterns for prediction and classification
- ❸ A software development framework
- ❹ A database management system

ANSWER: B

DETAILED EXPLANATION:

- ML is a subset of Artificial Intelligence (AI)
- Models are trained to recognize patterns in data
- Applications include prediction, classification, and risk management
- Techniques have evolved with advances in computing power and data availability

WHY OTHER OPTIONS ARE WRONG:

- **A:** ML is about learning patterns, not just storage
- **C:** ML is a set of techniques, not a software framework
- **D:** ML is not a database system

EXAM TRAP: ML = **training models to recognize patterns!**

QUESTION: Which is NOT an application of Machine Learning?

OPTIONS:

- ❶ Stock selection and trading
- ❷ Image recognition
- ❸ Manual data entry
- ❹ Fraud detection and credit scoring

ANSWER: C

DETAILED EXPLANATION:

- ML applications include stock selection, image recognition, game playing, autonomous vehicles
- Risk management: credit scoring, fraud detection
- Manual data entry is a human task, not an ML application

WHY OTHER OPTIONS ARE WRONG:

- **A:** Stock selection is a common ML application
- **B:** Image recognition is a major ML application
- **D:** Fraud detection and credit scoring are common ML applications

EXAM TRAP: ML applications include **prediction, classification, and pattern recognition!**

QUESTION: What advantage does ML offer over traditional econometrics?

OPTIONS:

- ❶ It always produces more interpretable results
- ❷ It can handle big data, non-linearity, and dimensionality reduction
- ❸ It requires no data cleaning
- ❹ It never overfits

ANSWER: B

DETAILED EXPLANATION:

- **Handling big data:** ML scales better with large datasets
- **Handling non-linearity:** ML captures complex relationships
- **Reducing dimensionality:** ML techniques like PCA help with many features
- **Handling missing data:** ML has flexible imputation methods

WHY OTHER OPTIONS ARE WRONG:

- **A:** ML models are often LESS interpretable
- **C:** ML requires extensive data cleaning
- **D:** ML models can overfit

EXAM TRAP: ML advantages = big data, non-linearity, dimensionality reduction!

QUESTION: How does ML differ from classical statistics in model selection?

OPTIONS:

- 1 ML relies on pre-specified theory; classical statistics lets data decide
- 2 ML lets data decide; classical statistics relies on pre-specified theory
- 3 Both rely equally on theory
- 4 Neither relies on theory

ANSWER: B

DETAILED EXPLANATION:

- **Classical statistics:** Theory-driven, pre-specified model
- **ML:** Data-driven, uses techniques like regularization
- **Classical:** Hypothesis testing and inference
- **ML:** Focus on out-of-sample prediction

WHY OTHER OPTIONS ARE WRONG:

- **A:** This reverses the roles
- **C:** They differ fundamentally
- **D:** Both rely on data, but differently

EXAM TRAP: ML = data-driven; Classical = theory-driven!

QUESTION: What is the primary focus of Machine Learning?

OPTIONS:

- ❶ Inference and hypothesis testing
- ❷ Out-of-sample prediction accuracy
- ❸ R-squared maximization
- ❹ Coefficient interpretation

ANSWER: B

DETAILED EXPLANATION:

- ML focuses on predictive performance
- Out-of-sample accuracy is the key metric
- Generalization to new data is paramount
- Less emphasis on inference (p-values, t-statistics)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Inference is the focus of classical statistics
- **C:** R-squared is a classical statistic
- **D:** Coefficient interpretation is secondary in ML

EXAM TRAP: ML focuses on **out-of-sample prediction!**

QUESTION: What is the ML term for "dependent variable"?

OPTIONS:

- ❶ Feature
- ❷ Target, output, or label
- ❸ Instance
- ❹ Attribute

ANSWER: B

DETAILED EXPLANATION:

- **Statistics:** Dependent variable, regressand
- **ML:** Target, output, label, response variable
- In classification: label indicates category
- In prediction: target is what we predict

WHY OTHER OPTIONS ARE WRONG:

- **A:** Feature is the independent variable
- **C:** Instance is a data point/observation
- **D:** Attribute is another term for feature

EXAM TRAP: Dependent variable in ML = **target, output, or label!**

Q7: ML Terminology (Independent Variable) - Tutorial

QUESTION: What is the ML term for "independent variable"?

OPTIONS:

- ❶ Output
- ❷ Label
- ❸ Feature, input, or signal
- ❹ Instance

ANSWER: C

DETAILED EXPLANATION:

- **Statistics:** Independent variable, predictor, regressor
- **ML:** Feature, input, signal, attribute
- Used to predict the target/output
- Can be numerical or categorical

WHY OTHER OPTIONS ARE WRONG:

- **A:** Output is the dependent variable
- **B:** Label is the dependent variable in classification
- **D:** Instance is a data point

EXAM TRAP: Independent variable in ML = feature, input, or signal!

Q8: ML Terminology (Estimation) - Tutorial

QUESTION: What is the ML term for "estimation"?

OPTIONS:

- ❶ Hypothesis testing
- ❷ Training or learning
- ❸ Model selection
- ❹ Validation

ANSWER: B

DETAILED EXPLANATION:

- **Statistics:** Estimation
- **ML:** Training or learning
- The estimator is called a learner or algorithm
- In classification: called a classifier

WHY OTHER OPTIONS ARE WRONG:

- **A:** Hypothesis testing is classical statistics
- **C:** Model selection is separate from estimation
- **D:** Validation is for model evaluation

EXAM TRAP: Estimation in ML = **training or learning!**

Q9: Data Point in ML - Tutorial

QUESTION: What is the ML term for a "data point"?

OPTIONS:

- ❶ Feature
- ❷ Label
- ❸ Instance or example
- ❹ Target

ANSWER: C

DETAILED EXPLANATION:

- **Statistics:** Observation, data point
- **ML:** Instance, example
- Each instance has features and possibly a label
- Represents one row in the dataset

WHY OTHER OPTIONS ARE WRONG:

- **A:** Feature is a column/variable
- **B:** Label is the target/outcome
- **D:** Target is the dependent variable

EXAM TRAP: Data point in ML = **instance or example!**

QUESTION: Match the ML term with its statistics equivalent: "Feature"

OPTIONS:

- ❶ Dependent variable
- ❷ Independent variable or predictor
- ❸ Data point
- ❹ Estimation

ANSWER: B

DETAILED EXPLANATION:

- **Statistics:** Independent variable, predictor, regressor
- **ML:** Feature, input, signal, attribute
- Used to explain/predict the target
- Can be continuous or categorical

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is Target/Label in ML
- **C:** This is Instance/Example in ML
- **D:** This is Training/Learning in ML

EXAM TRAP: Feature = **independent variable** in statistics!

QUESTION: What is unsupervised learning?

OPTIONS:

- ❶ Learning with labeled data to predict outcomes
- ❷ Learning with unlabeled data to recognize patterns
- ❸ Learning through trial and error with rewards
- ❹ Learning with partially labeled data

ANSWER: B

DETAILED EXPLANATION:

- No corresponding output/label for each observation
- Goal: recognize patterns, understand structure
- Applications: clustering, anomaly detection
- Examples: PCA, K-means clustering

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes supervised learning
- **C:** This describes reinforcement learning
- **D:** This describes semi-supervised learning

EXAM TRAP: Unsupervised = unlabeled data, pattern recognition!

QUESTION: Which is an application of unsupervised learning?

OPTIONS:

- ❶ Predicting house prices
- ❷ Clustering stocks with similar characteristics
- ❸ Classifying loan defaults
- ❹ Forecasting stock returns

ANSWER: B

DETAILED EXPLANATION:

- Clustering groups similar items without labels
- Other applications: anomaly detection, dimensionality reduction (PCA)
- Used to understand data structure
- Helps identify patterns before supervised learning

WHY OTHER OPTIONS ARE WRONG:

- **A:** Predicting house prices is supervised (regression)
- **C:** Classifying defaults is supervised (classification)
- **D:** Forecasting returns is supervised (time series prediction)

EXAM TRAP: Unsupervised = clustering and pattern discovery!

Q13: Supervised Learning Definition - Tutorial

QUESTION: What is supervised learning?

OPTIONS:

- ❶ Learning with unlabeled data
- ❷ Learning with labeled data to predict outcomes
- ❸ Learning through rewards and punishments
- ❹ Learning with partially labeled data

ANSWER: B

DETAILED EXPLANATION:

- Each observation has features AND a label/output
- Goal: predict target for new, unseen data
- Two types: prediction (continuous) and classification (categorical)
- Examples: linear regression, logistic regression

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes unsupervised learning
- **C:** This describes reinforcement learning
- **D:** This describes semi-supervised learning

EXAM TRAP: Supervised = labeled data, prediction/classification!

Q14: Prediction vs Classification - Tutorial

QUESTION: What is the difference between prediction and classification?

OPTIONS:

- ❶ Prediction uses categorical variables; classification uses continuous
- ❷ Prediction predicts numerical values; classification assigns categories
- ❸ Prediction is unsupervised; classification is supervised
- ❹ There is no difference

ANSWER: B

DETAILED EXPLANATION:

- **Prediction:** Numerical output (e.g., house price)
- **Classification:** Categorical output (e.g., default/no default)
- Both are forms of supervised learning
- Different evaluation metrics apply

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is reversed
- **C:** Both are supervised
- **D:** They are different types of supervised learning

EXAM TRAP: Prediction = numerical; Classification = categorical!

QUESTION: Which is an example of a classification problem?

OPTIONS:

- ❶ Forecasting GDP growth
- ❷ Predicting stock returns
- ❸ Determining if a loan will default
- ❹ Estimating house prices

ANSWER: C

DETAILED EXPLANATION:

- Classification: assigning to categories
- Loan default: Yes/No (binary classification)
- Target variable is categorical
- Other examples: fraud detection, credit rating

WHY OTHER OPTIONS ARE WRONG:

- **A:** GDP growth is numerical (prediction/regression)
- **B:** Stock returns are numerical (prediction)
- **D:** House prices are numerical (prediction)

EXAM TRAP: Classification = **categorical output** like default/no default!

Q16: Prediction Example - Tutorial

QUESTION: Which is an example of a prediction problem?

OPTIONS:

- ① Email spam detection
- ② Loan default classification
- ③ Predicting house sale prices
- ④ Image recognition

ANSWER: C

DETAILED EXPLANATION:

- Prediction: continuous numerical output
- House price: continuous numerical value
- Can be cross-sectional or time-series
- Examples: stock returns, GDP, revenue

WHY OTHER OPTIONS ARE WRONG:

- **A:** Email spam is classification (spam/not spam)
- **B:** Loan default is classification (default/no default)
- **D:** Image recognition is classification (object categories)

EXAM TRAP: Prediction = continuous numerical output!

QUESTION: What is semi-supervised learning?

OPTIONS:

- ❶ All data is labeled
- ❷ No data is labeled
- ❸ Only part of the data is labeled
- ❹ Learning through rewards and punishments

ANSWER: C

DETAILED EXPLANATION:

- Mix of labeled and unlabeled data
- Uses labeled data for supervised learning
- Uses unlabeled data to learn patterns in features
- Can improve performance with limited labeled data

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is supervised learning
- **B:** This is unsupervised learning
- **D:** This is reinforcement learning

EXAM TRAP: Semi-supervised = **partially labeled data!**

QUESTION: What is reinforcement learning?

OPTIONS:

- ❶ Learning from labeled examples
- ❷ Learning from unlabeled data
- ❸ Learning through trial and error with rewards
- ❹ Learning with partially labeled data

ANSWER: C

DETAILED EXPLANATION:

- No explicit labels
- Feedback through rewards/penalties
- Trial-and-error approach
- Output = recommended action given circumstances

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is supervised learning
- **B:** This is unsupervised learning
- **D:** This is semi-supervised learning

EXAM TRAP: Reinforcement = trial and error with rewards!

QUESTION: Which is an application of reinforcement learning in finance?

OPTIONS:

- ❶ Customer segmentation
- ❷ Optimal portfolio management
- ❸ Fraud detection
- ❹ House price prediction

ANSWER: B

DETAILED EXPLANATION:

- Reinforcement learning: making sequential decisions
- Portfolio management: buy/sell decisions over time
- Also: optimal trading, hedging, execution
- Learns from rewards (returns, risk reduction)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Customer segmentation is unsupervised
- **C:** Fraud detection is supervised classification
- **D:** House price prediction is supervised regression

EXAM TRAP: Reinforcement = **sequential decision making** like portfolio management!

QUESTION: What characterizes parametric machine learning methods?

OPTIONS:

- ❶ No assumptions about functional form
- ❷ Make assumptions about functional form between features and output
- ❸ Always outperform non-parametric methods
- ❹ Require no training data

ANSWER: B

DETAILED EXPLANATION:

- Assume a specific functional form (e.g., linear)
- Parameters are estimated from data
- Risk: wrong functional form = poor fit
- Examples: linear regression, logistic regression

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes non-parametric methods
- **C:** Not always true
- **D:** Parametric methods require training data

EXAM TRAP: Parametric = **assumes functional form!**

QUESTION: What characterizes non-parametric machine learning methods?

OPTIONS:

- ❶ Make assumptions about functional form
- ❷ No explicit assumption about functional form
- ❸ Require fewer observations
- ❹ Are always less accurate

ANSWER: B

DETAILED EXPLANATION:

- Flexible - no fixed functional form
- Capture complex patterns
- Require many observations for accuracy
- Examples: decision trees, KNN, neural networks

WHY OTHER OPTIONS ARE WRONG:

- **A:** This describes parametric methods
- **C:** Non-parametric typically requires MORE observations
- **D:** Can be very accurate with enough data

EXAM TRAP: Non-parametric = no fixed functional form!

QUESTION: What is a disadvantage of parametric methods?

OPTIONS:

- ❶ They are too flexible
- ❷ Risk of wrong functional form leading to poor fit
- ❸ They require too much data
- ❹ They are always slow

ANSWER: B

DETAILED EXPLANATION:

- Parametric methods assume a specific form
- If assumption is wrong, model fails to fit data
- Example: linear model for quadratic relationship
- Non-parametric methods avoid this risk

WHY OTHER OPTIONS ARE WRONG:

- **A:** Non-parametric are more flexible
- **C:** Non-parametric require more data
- **D:** Speed depends on implementation

EXAM TRAP: Parametric risk = **wrong functional form!**

QUESTION: What is a requirement for non-parametric methods?

OPTIONS:

- ❶ Few observations
- ❷ Many observations for accurate estimation
- ❸ No data at all
- ❹ Pre-specified functional form

ANSWER: B

DETAILED EXPLANATION:

- Non-parametric methods are data-intensive
- Need sufficient data to capture complex patterns
- More data = better estimation
- Parametric methods can work with less data

WHY OTHER OPTIONS ARE WRONG:

- **A:** Non-parametric needs **MANY** observations
- **C:** All methods need data
- **D:** This describes parametric methods

EXAM TRAP: Non-parametric = **needs many observations!**

QUESTION: What are the four main categories of machine learning?

OPTIONS:

- ❶ Supervised, Unsupervised, Semi-supervised, Reinforcement
- ❷ Training, Validation, Testing, Deployment
- ❸ Regression, Classification, Clustering, Dimensionality Reduction
- ❹ Parametric, Non-parametric, Linear, Non-linear

ANSWER: A

DETAILED EXPLANATION:

- **Supervised:** Labeled data, prediction/classification
- **Unsupervised:** Unlabeled data, pattern discovery
- **Semi-supervised:** Partially labeled data
- **Reinforcement:** Trial and error, rewards

WHY OTHER OPTIONS ARE WRONG:

- **B:** These are data splitting stages
- **C:** These are types of problems/tasks
- **D:** These are modeling approaches

EXAM TRAP: Four categories = Supervised, Unsupervised, Semi-supervised, Reinforcement!

QUESTION: Which ML type requires labeled data?

OPTIONS:

- ❶ Unsupervised learning
- ❷ Supervised learning
- ❸ Reinforcement learning
- ❹ Both supervised and semi-supervised

ANSWER: D

DETAILED EXPLANATION:

- Supervised: fully labeled data
- Semi-supervised: partially labeled data
- Unsupervised: no labels
- Reinforcement: no labels, only rewards

WHY OTHER OPTIONS ARE WRONG:

- **A:** Unsupervised has NO labels
- **B:** Correct but incomplete
- **C:** Reinforcement has no labels, only rewards

EXAM TRAP: Supervised and Semi-supervised = **labeled data!**

QUESTION: What is structured data?

OPTIONS:

- ❶ Data with no organization
- ❷ Data organized in rows and columns
- ❸ Data from images and videos
- ❹ Data in natural language

ANSWER: B

DETAILED EXPLANATION:

- Organized in rows and columns
- Each column = attribute/feature
- Each row = observation/instance
- Examples: census data, bank customer data

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is unstructured data
- **C:** This is unstructured data (images/videos)
- **D:** This is unstructured data (text)

EXAM TRAP: Structured data = rows and columns!

QUESTION: Which is an example of unstructured data?

OPTIONS:

- ❶ Census data
- ❷ Bank customer database
- ❸ Social media text posts
- ❹ Excel spreadsheet with numeric data

ANSWER: C

DETAILED EXPLANATION:

- Unstructured: not arranged in rows/columns
- Examples: text, images, audio, video
- Harder to analyze
- Requires pre-processing (NLP for text)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Census data is structured
- **B:** Bank database is structured
- **D:** Excel with numeric data is structured

EXAM TRAP: Unstructured = text, images, audio, video!

QUESTION: What is semi-structured data?

OPTIONS:

- ❶ Data with no structure at all
- ❷ Data with some structure but not fixed format
- ❸ Data in rows and columns
- ❹ Data that is always numeric

ANSWER: B

DETAILED EXPLANATION:

- Partly structured, partly unstructured
- Has some organization but not rigid
- Examples: emails, CSV files, HTML files
- Can contain both structured and unstructured elements

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is unstructured data
- **C:** This is structured data
- **D:** Can include text as well

EXAM TRAP: Semi-structured = **some structure**, not fixed format!

QUESTION: What is numerical data?

OPTIONS:

- ❶ Data with categories
- ❷ Data with numerical values that have natural ordering
- ❸ Data from images
- ❹ Data from text

ANSWER: B

DETAILED EXPLANATION:

- Continuous or discrete values
- Has natural ordering (\$50,000 > \$40,000)
- Can be used directly in ML models
- Examples: age, income, temperature

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is categorical data
- **C:** This is unstructured data
- **D:** This is unstructured data

EXAM TRAP: Numerical data = values with natural ordering!

QUESTION: What is categorical data?

OPTIONS:

- ❶ Data with continuous values
- ❷ Data that takes discrete values representing categories
- ❸ Data that is always numeric
- ❹ Data with no structure

ANSWER: B

DETAILED EXPLANATION:

- Takes discrete values
- Represents groups/categories
- Can be nominal (no order) or ordinal (ordered)
- Examples: marital status, gender, region

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is numerical data
- **C:** Categorical can be non-numeric
- **D:** This is unstructured data

EXAM TRAP: Categorical data = discrete categories!

QUESTION: What is nominal data?

OPTIONS:

- ❶ Categorical data with natural ordering
- ❷ Categorical data with no natural ordering
- ❸ Continuous numerical data
- ❹ Time-series data

ANSWER: B

DETAILED EXPLANATION:

- Categories with NO inherent order
- Example: marital status, gender, eye color
- Cannot be used as numeric without encoding
- One-hot encoding is used

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is ordinal data
- **C:** This is numerical data
- **D:** This is longitudinal data

EXAM TRAP: Nominal = **no natural ordering!**

QUESTION: What is ordinal data?

OPTIONS:

- ❶ Categorical data with natural ordering
- ❷ Categorical data with no natural ordering
- ❸ Continuous numerical data
- ❹ Binary data

ANSWER: A

DETAILED EXPLANATION:

- Categories have natural ordering
- Intervals between values not uniform
- Examples: educational level, credit ratings, satisfaction scores
- Can be encoded with ordinal values

WHY OTHER OPTIONS ARE WRONG:

- **B:** This is nominal data
- **C:** This is numerical data
- **D:** Binary is a special case

EXAM TRAP: Ordinal = natural ordering of categories!

QUESTION: Which is an example of ordinal data?

OPTIONS:

- ① Gender (Male/Female)
- ② Marital status (Single/Married/Divorced)
- ③ Credit ratings (AAA, AA, A, BBB)
- ④ Country of origin

ANSWER: C

DETAILED EXPLANATION:

- Ordinal: categories have natural ordering
- Credit ratings: AAA $\hat{<}$ AA $\hat{<}$ A $\hat{<}$ BBB
- Ordering is meaningful
- Intervals between ratings are not equal

WHY OTHER OPTIONS ARE WRONG:

- **A:** Gender is nominal (no order)
- **B:** Marital status is nominal (no order)
- **D:** Country is nominal (no order)

EXAM TRAP: Credit ratings = ordinal data!

QUESTION: What is binary data?

OPTIONS:

- ❶ Data with two possible values
- ❷ Data with multiple categories
- ❸ Continuous data
- ❹ Text data

ANSWER: A

DETAILED EXPLANATION:

- Special case of categorical data
- Only two possible values (0/1, Yes/No)
- Example: default/no default, fraud/legitimate
- Can be used directly in ML models

WHY OTHER OPTIONS ARE WRONG:

- **B:** This is multi-category categorical
- **C:** Binary is not continuous
- **D:** Binary is not text

EXAM TRAP: Binary data = two possible values!

QUESTION: What is cross-sectional data?

OPTIONS:

- ❶ Data spanning multiple time periods
- ❷ Data for a single point in time
- ❸ Data connected spatially
- ❹ Data from networks

ANSWER: B

DETAILED EXPLANATION:

- Snapshot at a specific moment
- Observations are independent
- Order does not matter
- Examples: loan portfolio characteristics, survey responses

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is longitudinal/time-series data
- **C:** This is spatial data
- **D:** This is network data

EXAM TRAP: Cross-sectional = **single point in time!**

QUESTION: What is longitudinal data?

OPTIONS:

- ❶ Data for a single point in time
- ❷ Data spanning multiple time periods or connected spatially
- ❸ Data that is always cross-sectional
- ❹ Data that cannot be analyzed

ANSWER: B

DETAILED EXPLANATION:

- Observations are related temporally or spatially
- Time-series is the most common example
- Order matters in analysis
- Examples: stock prices, economic data

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is cross-sectional data
- **C:** Longitudinal is NOT cross-sectional
- **D:** Longitudinal data can be analyzed

EXAM TRAP: Longitudinal = **multiple time periods or spatial connections!**

QUESTION: What is time-series data?

OPTIONS:

- ❶ Data measured at a single point in time
- ❷ Data measured over time with chronological order
- ❸ Data from different locations
- ❹ Data with no ordering

ANSWER: B

DETAILED EXPLANATION:

- Observations over time
- Chronological order matters
- Observations are not independent
- Examples: stock prices, GDP, yields

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is cross-sectional data
- **C:** This is spatial data
- **D:** Time-series has ordering

EXAM TRAP: Time-series = data over time with chronological order!

QUESTION: What are the four scales of data measurement?

OPTIONS:

- ❶ Nominal, Ordinal, Interval, Ratio
- ❷ Binary, Ordinal, Interval, Ratio
- ❸ Nominal, Binary, Ordinal, Continuous
- ❹ Categorical, Numerical, Text, Image

ANSWER: A

DETAILED EXPLANATION:

- **Nominal:** Categories without order
- **Ordinal:** Categories with order
- **Interval:** Equal intervals, no true zero
- **Ratio:** Equal intervals, true zero exists

WHY OTHER OPTIONS ARE WRONG:

- **B:** Missing Nominal
- **C:** Missing Interval and Ratio
- **D:** These are data types, not measurement scales

EXAM TRAP: Four scales = **Nominal, Ordinal, Interval, Ratio!**

QUESTION: What distinguishes interval from ratio data?

OPTIONS:

- ❶ Ratio data has no true zero
- ❷ Interval data has a true zero
- ❸ Ratio data has a true zero; interval does not
- ❹ There is no difference

ANSWER: C

DETAILED EXPLANATION:

- **Interval:** No true zero (e.g., temperature in °C)
- **Ratio:** True zero exists (e.g., height, income)
- Ratio: 0 means absence of the quantity
- Interval: 0 is arbitrary (0°C doesn't mean no temperature)

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is reversed
- **B:** This is reversed
- **D:** There is a difference (true zero)

EXAM TRAP: Ratio = true zero; Interval = no true zero!

QUESTION: Which data type has a true zero point?

OPTIONS:

- ① Nominal
- ② Ordinal
- ③ Interval
- ④ Ratio

ANSWER: D

DETAILED EXPLANATION:

- Ratio scale: true zero exists
- 0 means absence of the quantity
- Examples: height, weight, income, distance
- Allows for meaningful ratios (e.g., 2x as tall)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Nominal has no order or zero
- **B:** Ordinal has order but no true zero
- **C:** Interval has no true zero

EXAM TRAP: Ratio = true zero!

QUESTION: What is Exploratory Data Analysis (EDA)?

OPTIONS:

- ❶ Building the final model
- ❷ The process of collecting, cleaning, visualizing, and analyzing data before modeling
- ❸ Deploying the model to production
- ❹ Testing the model on new data

ANSWER: B

DETAILED EXPLANATION:

- Essential first step in ML projects
- Includes data collection, cleaning, visualization
- Understand data structure and relationships
- Helps select features for modeling

WHY OTHER OPTIONS ARE WRONG:

- **A:** EDA is before model building
- **C:** This is deployment
- **D:** This is testing/evaluation

EXAM TRAP: EDA = **exploring data BEFORE modeling!**

QUESTION: What percentage of time do data analysts often spend cleaning data?

OPTIONS:

- ☒ 1 10%
- ☐ 2 30%
- ☐ 3 Up to 80%
- ☐ 4 100%

ANSWER: C

DETAILED EXPLANATION:

- Data cleaning is time-consuming
- Often up to 80% of project time
- Essential for successful ML projects
- "Garbage in, garbage out" (GIGO)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Much more than 10%
- **B:** Typical is 80%, not 30%
- **D:** Not 100%, but very significant

EXAM TRAP: Data cleaning = up to 80% of project time!

QUESTION: What does GIGO stand for?

OPTIONS:

- ❶ Good Inputs, Great Outputs
- ❷ Garbage In, Garbage Out
- ❸ Great Inputs, Great Outputs
- ❹ Graphics In, Graphics Out

ANSWER: B

DETAILED EXPLANATION:

- Analysis quality depends on data quality
- Poor data leads to poor results
- Emphasizes importance of data cleaning
- Fundamental principle in data science

WHY OTHER OPTIONS ARE WRONG:

- A: Incorrect expansion
- C: Incorrect expansion
- D: Incorrect expansion

EXAM TRAP: GIGO = Garbage In, Garbage Out!

QUESTION: What is NOT a reason for data cleaning?

OPTIONS:

- ❶ Inconsistent recording
- ❷ Duplicate observations
- ❸ Outliers
- ❹ Increasing data size

ANSWER: D

DETAILED EXPLANATION:

- Data cleaning removes errors, not increases data
- Reasons: inconsistent recording, duplicates, outliers, missing data
- Goal: improve data quality
- Data augmentation increases data size (separate technique)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Inconsistent recording IS a reason for cleaning
- **B:** Duplicates IS a reason for cleaning
- **C:** Outliers IS a reason for cleaning

EXAM TRAP: Data cleaning = quality improvement, not size increase!

QUESTION: What is an outlier in a dataset?

OPTIONS:

- ❶ The average value in the dataset
- ❷ Observations significantly different from the rest of the data
- ❸ The most common value in the dataset
- ❹ Data that is always correct

ANSWER: B

DETAILED EXPLANATION:

- Observations several standard deviations from mean
- May be errors or genuine rare events
- Can significantly affect results
- Treatment depends on application

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is the mean
- **C:** This is the mode
- **D:** Outliers may be errors

EXAM TRAP: Outliers = significantly different observations!

QUESTION: How should outliers be treated?

OPTIONS:

- ① Always remove them
- ② Always keep them
- ③ Depends on the specific application and context
- ④ They are never a problem

ANSWER: C

DETAILED EXPLANATION:

- Sometimes errors → remove
- Sometimes genuine rare events → keep
- Fraud detection: outliers may be valuable
- No one-size-fits-all solution

WHY OTHER OPTIONS ARE WRONG:

- **A:** Not always remove
- **B:** Not always keep
- **D:** Outliers ARE a problem to handle

EXAM TRAP: Outlier treatment = **depends on application!**

QUESTION: Which models are robust to outliers?

OPTIONS:

- ❶ Linear regression
- ❷ Tree-based models (decision trees, random forests)
- ❸ Logistic regression
- ❹ All models are equally sensitive

ANSWER: B

DETAILED EXPLANATION:

- Tree-based models split data into regions
- Outliers get their own branches
- Less affected by extreme values
- Chapter 4 covers these models

WHY OTHER OPTIONS ARE WRONG:

- **A:** Linear regression is sensitive to outliers
- **C:** Logistic regression can be sensitive
- **D:** Models have different sensitivities

EXAM TRAP: Tree-based models = robust to outliers!

QUESTION: What is imputation in data cleaning?

OPTIONS:

- ❶ Removing all rows with missing data
- ❷ Replacing missing values with estimated values
- ❸ Ignoring missing values
- ❹ Creating more missing values

ANSWER: B

DETAILED EXPLANATION:

- Replace missing observations with estimates
- Common methods: mean, median, mode imputation
- More sophisticated: model-based imputation
- Alternative to deleting observations

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is deletion, not imputation
- **C:** Ignoring is not a solution
- **D:** Imputation FILLS missing values

EXAM TRAP: Imputation = replacing missing values with estimates!

QUESTION: What is informative missingness?

OPTIONS:

- ❶ Missing data that is random
- ❷ Missing data that is associated with the outcome
- ❸ Missing data that is always structural
- ❹ Missing data that cannot be imputed

ANSWER: B

DETAILED EXPLANATION:

- Pattern of missing data relates to outcome
- Can induce significant bias
- Example: credit card data only for solvent clients
- Requires careful handling

WHY OTHER OPTIONS ARE WRONG:

- **A:** Random missingness is not informative
- **C:** Structural missing is different
- **D:** It can be handled carefully

EXAM TRAP: Informative missingness = missing pattern related to outcome!

QUESTION: Which is a data visualization tool?

OPTIONS:

- ① Histogram
- ② Correlation matrix
- ③ Scatter plot
- ④ All of the above

ANSWER: D

DETAILED EXPLANATION:

- **Histogram:** Distribution of a single variable
- **Scatter plot:** Relationship between two variables
- **Correlation matrix:** Relationships among multiple variables
- **Boxplot:** Summary of distribution with outliers

WHY OTHER OPTIONS ARE WRONG:

- **A:** Correct but incomplete
- **B:** Correct but incomplete
- **C:** Correct but incomplete

EXAM TRAP: All are data visualization tools!

Q51: Histogram Purpose - Tutorial

QUESTION: What does a histogram show?

OPTIONS:

- ❶ Relationship between two variables
- ❷ The distribution/shape of a single variable
- ❸ The correlation between features
- ❹ The missing values in the data

ANSWER: B

DETAILED EXPLANATION:

- Count of observations in bins
- Shows distribution shape
- Identifies skewness, outliers, patterns
- Area under histogram equals 1 (when density)

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is scatter plot
- **C:** This is correlation matrix
- **D:** This is not what histogram shows

EXAM TRAP: Histogram shows **distribution of one variable!**

QUESTION: What is skewness in a distribution?

OPTIONS:

- ① Symmetric distribution
- ② Asymmetric distribution (longer tail on one side)
- ③ Distribution with no tails
- ④ Distribution with multiple peaks

ANSWER: B

DETAILED EXPLANATION:

- **Negative skew:** Longer left tail
- **Positive skew:** Longer right tail
- Example: stock returns have negative skew
- Can be addressed with transformations (log)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Symmetric is no skew
- **C:** All distributions have tails
- **D:** This is multimodal

EXAM TRAP: Skewness = **asymmetric** distribution!

QUESTION: What is a Q-Q plot used for?

OPTIONS:

- ① Showing correlations
- ② Comparing empirical distribution to theoretical distribution
- ③ Showing missing values
- ④ Showing outliers only

ANSWER: B

DETAILED EXPLANATION:

- Q-Q = Quantile-Quantile
- Compares empirical vs theoretical quantiles
- Points on identity line = distribution matches
- Normal Q-Q plot: checks normality

WHY OTHER OPTIONS ARE WRONG:

- **A:** Correlation is different
- **C:** Missing values are different
- **D:** Shows more than just outliers

EXAM TRAP: Q-Q plot = empirical vs theoretical distribution!

QUESTION: What does a boxplot show?

OPTIONS:

- ❶ Mean and standard deviation only
- ❷ Five-number summary and outliers
- ❸ Correlation between variables
- ❹ The relationship between two variables

ANSWER: B

DETAILED EXPLANATION:

- Box = 1st quartile to 3rd quartile
- Line inside = median
- Whiskers = $1.5 \times \text{IQR}$
- Points beyond whiskers = outliers

WHY OTHER OPTIONS ARE WRONG:

- **A:** Boxplot shows quartiles, not just mean/sd
- **C:** Correlation is different
- **D:** Scatter plot shows relationships

EXAM TRAP: Boxplot = five-number summary + outliers!

QUESTION: What does a correlation matrix show?

OPTIONS:

- ❶ Distribution of each variable
- ❷ Pairwise correlations between variables
- ❸ Missing values
- ❹ Outliers

ANSWER: B

DETAILED EXPLANATION:

- Matrix of pairwise correlations
- Values between -1 and 1
- Diagonal = 1 (variable with itself)
- Helps identify highly correlated features

WHY OTHER OPTIONS ARE WRONG:

- **A:** Histogram shows distribution
- **C:** Missing values are different
- **D:** Outliers are different

EXAM TRAP: Correlation matrix = pairwise correlations!

QUESTION: What is the formula for standardization?

OPTIONS:

1 $\tilde{x}_{ij} = \frac{x_{ij} - \min(x_i)}{\max(x_i) - \min(x_i)}$

2 $\tilde{x}_{ij} = \frac{x_{ij} - \hat{\mu}_i}{\hat{\sigma}_i}$

3 $\tilde{x}_{ij} = x_{ij} - \hat{\mu}_i$

4 $\tilde{x}_{ij} = \frac{x_{ij}}{\hat{\sigma}_i}$

ANSWER: B

DETAILED EXPLANATION:

- $\hat{\mu}_i$: sample mean of variable i
- $\hat{\sigma}_i$: sample standard deviation
- New variable: mean = 0, variance = 1
- Can take any value (positive/negative)

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is normalization (min-max)
- **C:** Only subtracts mean (centering)
- **D:** Only divides by standard deviation

EXAM TRAP: Standardization = $(x - \text{mean})/sd$!

QUESTION: What is the formula for normalization (min-max transformation)?

OPTIONS:

1 $\tilde{x}_{ij} = \frac{x_{ij} - \min(x_i)}{\max(x_i) - \min(x_i)}$

2 $\tilde{x}_{ij} = \frac{x_{ij} - \hat{\mu}_i}{\hat{\sigma}_i}$

3 $\tilde{x}_{ij} = \frac{x_{ij}}{\hat{\mu}_i}$

4 $\tilde{x}_{ij} = x_{ij} - \hat{\mu}_i$

ANSWER: A

DETAILED EXPLANATION:

- Bounds between 0 and 1
- Uses min and max of the variable
- All values in [0, 1] range
- Mean not necessarily 0, variance not necessarily 1

WHY OTHER OPTIONS ARE WRONG:

- **B:** This is standardization
- **C:** Not a standard method
- **D:** This is centering only

EXAM TRAP: Normalization = $(x - \min)/(\max - \min)$!

QUESTION: When is standardization preferred over normalization?

OPTIONS:

- ❶ When data has no outliers
- ❷ When data contains outliers
- ❸ When data is categorical
- ❹ When data is already in $[0,1]$ range

ANSWER: B

DETAILED EXPLANATION:

- Standardization handles outliers better
- Normalization squashes data into tight range
- Outliers can distort min-max range
- Standardization uses mean and sd (more robust)

WHY OTHER OPTIONS ARE WRONG:

- **A:** Normalization works without outliers
- **C:** Both are for numerical data
- **D:** Scaling not needed if already $[0,1]$

EXAM TRAP: Standardization = better for data with outliers!

QUESTION: What is NOT a reason for scaling data?

OPTIONS:

- ❶ Numerical stability of learning algorithm
- ❷ Ease of interpretation of parameter estimates
- ❸ To make data categorical
- ❹ To check if predictions are extrapolations

ANSWER: C

DETAILED EXPLANATION:

- Scaling doesn't make data categorical
- Reasons: numerical stability, interpretation, extrapolation detection
- Scaling transforms numerical values
- Categorical variables need encoding (not scaling)

WHY OTHER OPTIONS ARE WRONG:

- **A:** This IS a reason for scaling
- **B:** This IS a reason for scaling
- **D:** This IS a reason for scaling

EXAM TRAP: Scaling = numerical transformation, not categorization!

QUESTION: Why apply a log transformation to skewed data?

OPTIONS:

- ❶ To make data categorical
- ❷ To reduce skewness and make distribution more symmetric
- ❸ To increase skewness
- ❹ To remove all outliers

ANSWER: B

DETAILED EXPLANATION:

- Log transformation reduces positive skew
- Data becomes better behaved
- Example: trading volume distribution becomes more symmetric
- Rule of thumb: if $\text{max}/\text{min} > 10$, consider log transform

WHY OTHER OPTIONS ARE WRONG:

- **A:** Log doesn't make data categorical
- **C:** Log REDUCES skewness
- **D:** Log doesn't remove outliers

EXAM TRAP: Log transform = **reduce skewness!**

QUESTION: What is feature extraction/engineering?

OPTIONS:

- ❶ Removing all features
- ❷ Transforming raw attributes into features suitable for analysis
- ❸ Always using raw data without modification
- ❹ Only using categorical features

ANSWER: B

DETAILED EXPLANATION:

- Process of creating useful features from raw data
- Includes encoding categorical variables
- May include creating derived features
- Critical step in ML pipeline

WHY OTHER OPTIONS ARE WRONG:

- **A:** Feature extraction creates, not removes
- **C:** Raw data often needs transformation
- **D:** Both numerical and categorical features used

EXAM TRAP: Feature extraction = transforming raw attributes into features!

QUESTION: What is one-hot encoding?

OPTIONS:

- ❶ Converting numerical data to categorical
- ❷ Creating binary dummy variables for each category
- ❸ Removing categorical variables
- ❹ Converting text to numbers

ANSWER: B

DETAILED EXPLANATION:

- Also called binarization
- Each category becomes a binary (0/1) variable
- Only one dummy = 1 for each observation
- Essential for nominal categorical variables

WHY OTHER OPTIONS ARE WRONG:

- **A:** One-hot encodes categorical to binary
- **C:** It encodes, not removes
- **D:** Text encoding is different (NLP)

EXAM TRAP: One-hot encoding = **binary dummies for each category!**

Q63: One-Hot Encoding Example - Tutorial

QUESTION: How would you encode "Marital Status" (Single/Married/Divorced)?

OPTIONS:

- ❶ Single=0, Married=1, Divorced=2
- ❷ Create three binary dummies: Single, Married, Divorced
- ❸ Remove the variable
- ❹ Convert to text

ANSWER: B

DETAILED EXPLANATION:

- Marital status is nominal (no ordering)
- One-hot: three binary variables
- Single: [1,0,0], Married: [0,1,0], Divorced: [0,0,1]
- Avoids implying ordering between categories

WHY OTHER OPTIONS ARE WRONG:

- **A:** This implies ordering (incorrect for nominal)
- **C:** Removing loses valuable information
- **D:** This doesn't create usable features

EXAM TRAP: Nominal data = one-hot encoding (not numeric assignment)!

Q64: Dummy Variable Trap - Tutorial

QUESTION: What is the dummy variable trap?

OPTIONS:

- ❶ Using too many dummy variables
- ❷ Creating perfect multicollinearity when including all dummies and intercept
- ❸ Not including enough dummy variables
- ❹ Using only one dummy variable

ANSWER: B

DETAILED EXPLANATION:

- Including all dummies + intercept creates perfect collinearity
- Example: $\text{Male} + \text{Female} + \text{Intercept} = 1$ for all observations
- Solution: omit one dummy or set intercept to zero
- Also known as the reference category approach

WHY OTHER OPTIONS ARE WRONG:

- **A:** Too many is not the trap
- **C:** Not enough is not the trap
- **D:** One dummy is fine

EXAM TRAP: Dummy variable trap = perfect multicollinearity with intercept!

QUESTION: How can the dummy variable trap be avoided?

OPTIONS:

- ❶ Include all dummies and intercept
- ❷ Omit one dummy variable or set intercept to zero
- ❸ Always use two dummies only
- ❹ Never use dummy variables

ANSWER: B

DETAILED EXPLANATION:

- **Omit one dummy:** Reference category
- **Set intercept to zero:** Include all dummies
- Both solutions avoid perfect multicollinearity
- Most common: omit one dummy variable

WHY OTHER OPTIONS ARE WRONG:

- **A:** This creates the trap
- **C:** May not be sufficient
- **D:** Dummies are often needed

EXAM TRAP: Solutions = omit one dummy or set intercept to zero!

QUESTION: How should ordinal categorical data be encoded?

OPTIONS:

- ❶ Always use one-hot encoding
- ❷ Use numeric values that preserve the ordering
- ❸ Remove the variable
- ❹ Convert to text

ANSWER: B

DETAILED EXPLANATION:

- Ordinal has natural ordering
- Assign values: 0, 1, 2, ... preserving order
- Example: Education (0=HS, 1=College, 2=Post-grad)
- Can be used as feature in models

WHY OTHER OPTIONS ARE WRONG:

- **A:** One-hot loses ordering information
- **C:** Removing loses valuable information
- **D:** Text is not usable directly

EXAM TRAP: Ordinal = numeric encoding preserving order!

QUESTION: What is discretization (binning)?

OPTIONS:

- ❶ Converting continuous data to categorical data by grouping into ranges
- ❷ Converting categorical to numerical
- ❸ Removing outliers
- ❹ Increasing data size

ANSWER: A

DETAILED EXPLANATION:

- Also called binning
- Group continuous values into categories
- Example: Firm size by market cap ranges
- Useful when exact values don't matter

WHY OTHER OPTIONS ARE WRONG:

- **B:** This is encoding, not discretization
- **C:** Outliers are different
- **D:** Discretization doesn't increase size

EXAM TRAP: Discretization = continuous → categorical via binning!

QUESTION: When would you discretize a continuous variable?

OPTIONS:

- ❶ When exact values are important
- ❷ When you only care about ranges (e.g., capital above/below threshold)
- ❸ Always discretize all variables
- ❹ Never discretize continuous variables

ANSWER: B

DETAILED EXPLANATION:

- Example: Regulatory capital - above/below threshold
- Or market cap ranges (Micro, Small, Medium, Large)
- Useful when categories are meaningful
- Loss of information if exact values matter

WHY OTHER OPTIONS ARE WRONG:

- **A:** If exact values matter, keep continuous
- **C:** Should only discretize when meaningful
- **D:** Sometimes discretization is useful

EXAM TRAP: Discretize when **categories are more meaningful than exact values!**

QUESTION: Which is NOT a method of feature extraction?

OPTIONS:

- ❶ One-hot encoding categorical variables
- ❷ Applying a log transformation
- ❸ Discretizing continuous variables
- ❹ Deleting all features

ANSWER: D

DETAILED EXPLANATION:

- Feature extraction creates/transforms features
- One-hot encoding: categorical to binary
- Log transformation: numerical to reduce skew
- Discretization: continuous to categorical
- Deleting features is feature selection, not extraction

WHY OTHER OPTIONS ARE WRONG:

- **A:** This IS feature extraction
- **B:** This IS feature extraction
- **C:** This IS feature extraction

EXAM TRAP: Feature extraction = **creating/transforming**, not deleting!

QUESTION: Which encoding method is appropriate for nominal data?

OPTIONS:

- ① Ordinal encoding (0,1,2,...)
- ② One-hot encoding (binary dummies)
- ③ Log transformation
- ④ Standardization

ANSWER: B

DETAILED EXPLANATION:

- Nominal data has no natural ordering
- One-hot encoding: separate binary variable for each category
- Avoids implying false ordering
- Examples: marital status, gender, region

WHY OTHER OPTIONS ARE WRONG:

- **A:** Ordinal encoding assumes ordering (wrong for nominal)
- **C:** Log is for numerical data
- **D:** Standardization is for numerical data

EXAM TRAP: Nominal data = one-hot encoding!

QUESTION: What is Principal Components Analysis (PCA)?

OPTIONS:

- ❶ A supervised learning method
- ❷ A dimensionality reduction technique that finds linear combinations of original features
- ❸ A clustering algorithm
- ❹ A method for handling missing data

ANSWER: B

DETAILED EXPLANATION:

- Most common dimensionality reduction technique
- Finds linear combinations of original predictors
- Summarizes variability in data
- Principal components are uncorrelated

WHY OTHER OPTIONS ARE WRONG:

- **A:** PCA is UNSUPERVISED learning
- **C:** PCA is for dimensionality reduction, not clustering
- **D:** Missing data handling is different

EXAM TRAP: PCA = **unsupervised dimensionality reduction!**

QUESTION: What is the primary purpose of PCA?

OPTIONS:

- ❶ To increase the number of features
- ❷ To reduce dimensionality while preserving variability
- ❸ To classify observations
- ❹ To cluster data

ANSWER: B

DETAILED EXPLANATION:

- Reduce number of features (dimensions)
- Preserve most variability in data
- Components ordered by explained variance
- First PC explains most variance

WHY OTHER OPTIONS ARE WRONG:

- **A:** PCA REDUCES features
- **C:** PCA is not classification
- **D:** PCA is not clustering

EXAM TRAP: PCA = dimensionality reduction, not classification/clustering!

QUESTION: What is the first principal component?

OPTIONS:

- ❶ The linear combination that captures the most variability in the data
- ❷ The average of all features
- ❸ The first feature in the dataset
- ❹ The feature with the highest value

ANSWER: A

DETAILED EXPLANATION:

- Linear combination of original predictors
- Captures most variability among all possible combinations
- Maximizes variance
- Component weights show feature importance

WHY OTHER OPTIONS ARE WRONG:

- **B:** Average is not a principal component
- **C:** First PC is a combination, not a single feature
- **D:** PC is not about values, about variability

EXAM TRAP: First PC = captures most variability!

QUESTION: What does it mean that principal components are orthogonal?

OPTIONS:

- ❶ They are highly correlated
- ❷ They are uncorrelated (independent)
- ❸ They have equal variance
- ❹ They have equal means

ANSWER: B

DETAILED EXPLANATION:

- Orthogonal = uncorrelated
- Each PC captures different aspect of variability
- No overlap in information
- Useful for avoiding multicollinearity

WHY OTHER OPTIONS ARE WRONG:

- **A:** Orthogonal means NOT correlated
- **C:** PCs have decreasing variance
- **D:** Means are not necessarily equal

EXAM TRAP: Orthogonal = **uncorrelated**!

QUESTION: What is the formula for a principal component?

OPTIONS:

- ① $PC_j = b_{1j}P_1 + b_{2j}P_2 + \dots + b_{mj}P_m$
- ② $PC_j = \frac{1}{m} \sum_{i=1}^m P_i$
- ③ $PC_j = \max(P_1, P_2, \dots, P_m)$
- ④ $PC_j = P_1 \times P_2 \times \dots \times P_m$

ANSWER: A

DETAILED EXPLANATION:

- P_i : original predictors (standardized)
- b_{ij} : component weights
- Linear combination of original features
- Weights show importance of each feature

WHY OTHER OPTIONS ARE WRONG:

- **B:** This is the mean
- **C:** This is the maximum
- **D:** This is the product

EXAM TRAP: PC = linear combination of features with weights!

QUESTION: What do component weights (b_{ij}) represent in PCA?

OPTIONS:

- ❶ The mean of each feature
- ❷ The importance of each feature for the principal component
- ❸ The standard deviation of each feature
- ❹ The correlation between features

ANSWER: B

DETAILED EXPLANATION:

- Weights indicate feature importance
- Larger absolute weight = more important
- Positive weights: feature moves with PC
- Negative weights: feature moves opposite to PC

WHY OTHER OPTIONS ARE WRONG:

- **A:** Means are different
- **C:** Standard deviations are different
- **D:** Correlations are different

EXAM TRAP: Weights = feature importance for each PC!

QUESTION: In the Treasury yield curve PCA, what does PC1 represent?

OPTIONS:

- ① Slope of the yield curve
- ② Level of the yield curve
- ③ Twist of the yield curve
- ④ Curvature of the yield curve

ANSWER: B

DETAILED EXPLANATION:

- PC1 loads positively on all maturities
- All yields move in same direction
- Represents parallel shift (level)
- Explains 82% of variation

WHY OTHER OPTIONS ARE WRONG:

- **A:** Slope is PC2 (short/long opposite)
- **C:** Twist is PC3
- **D:** Curvature is not mentioned

EXAM TRAP: PC1 = level of yield curve!

QUESTION: In the Treasury yield curve PCA, what does PC2 represent?

OPTIONS:

- ❶ Level of the yield curve
- ❷ Slope of the yield curve
- ❸ Twist of the yield curve
- ❹ Curvature of the yield curve

ANSWER: B

DETAILED EXPLANATION:

- PC2 loads negatively on short-term yields
- Loads positively on long-term yields
- Represents steepening/flattening (slope)
- Short and long yields move opposite

WHY OTHER OPTIONS ARE WRONG:

- **A:** Level is PC1
- **C:** Twist is PC3
- **D:** Curvature is not mentioned

EXAM TRAP: PC2 = slope of yield curve!

QUESTION: In the Treasury yield curve PCA, what does PC3 represent?

OPTIONS:

- ① Level of the yield curve
- ② Slope of the yield curve
- ③ Twist of the yield curve
- ④ The overall average

ANSWER: C

DETAILED EXPLANATION:

- PC3 loads positively on 1-year and 20+ year
- Loads negatively on 2-10 year yields
- Represents butterfly/twist
- Captures non-parallel shifts

WHY OTHER OPTIONS ARE WRONG:

- **A:** Level is PC1
- **B:** Slope is PC2
- **D:** Not the average

EXAM TRAP: PC3 = twist of yield curve!

QUESTION: In the yield curve example, how much variance do the first 3 PCs explain?

OPTIONS:

- ☒ 1 50%
- ☐ 2 75%
- ☐ 3 More than 98%
- ☐ 4 100%

ANSWER: C

DETAILED EXPLANATION:

- PC1: 82%
- PC2: ~12% (to reach 94%)
- PC3: ~4% (to reach 98+%)
- Total variance explained by first 3 PCs > 98%

WHY OTHER OPTIONS ARE WRONG:

- **A:** More than 50%
- **B:** More than 75%
- **D:** Not 100% (some in later PCs)

EXAM TRAP: First 3 PCs explain >98% of variance in yield curve!

QUESTION: How is the number of PCs chosen using an elbow plot?

OPTIONS:

- ❶ Choose all PCs
- ❷ Choose the point where explained variance drops off (elbow)
- ❸ Always choose the first 5 PCs
- ❹ Choose the PC with highest variance

ANSWER: B

DETAILED EXPLANATION:

- Plot: PC number vs variance explained
- Elbow: point where marginal variance drops
- Keep PCs before elbow
- After elbow: diminishing returns

WHY OTHER OPTIONS ARE WRONG:

- **A:** Too many PCs may be unnecessary
- **C:** Optimal number varies by dataset
- **D:** Need to consider multiple PCs

EXAM TRAP: Elbow = point where variance drops off!

QUESTION: How does PCA help with multicollinearity?

OPTIONS:

- ❶ It creates correlated features
- ❷ It creates uncorrelated features (orthogonal PCs)
- ❸ It removes all features
- ❹ It increases the number of features

ANSWER: B

DETAILED EXPLANATION:

- Original features may be highly correlated
- PCs are constructed to be uncorrelated
- Reduces multicollinearity problems
- Improves numerical stability of models

WHY OTHER OPTIONS ARE WRONG:

- **A:** PCs are uncorrelated, not correlated
- **C:** PCA doesn't remove all features
- **D:** PCA reduces features

EXAM TRAP: PCA creates **uncorrelated features** (solves multicollinearity)!

QUESTION: What should be done before applying PCA?

OPTIONS:

- ❶ Nothing - PCA handles everything
- ❷ Scale/standardize the features
- ❸ Remove all categorical variables
- ❹ Add more features

ANSWER: B

DETAILED EXPLANATION:

- Features should be scaled (mean 0, variance 1)
- Prevents features with large scales from dominating
- Remove skewness before scaling
- PCA is sensitive to scale differences

WHY OTHER OPTIONS ARE WRONG:

- **A:** Scaling is necessary
- **C:** Categorical variables need encoding first
- **D:** Adding features defeats the purpose

EXAM TRAP: Scale/standardize features before PCA!

QUESTION: How did Bank of Jamaica use PCA?

OPTIONS:

- ❶ For customer segmentation
- ❷ To develop a VaR methodology for interest rate risk
- ❸ For fraud detection
- ❹ For image recognition

ANSWER: B

DETAILED EXPLANATION:

- Developed PCA-based VaR methodology
- Assessed interest rate risk of Jamaican banks
- Decomposed yield curve movements into orthogonal factors
- More comprehensive than traditional VaR methods

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is a different application
- **C:** Fraud detection is different
- **D:** Image recognition is different

EXAM TRAP: BOJ used PCA for [interest rate risk VaR](#)!

QUESTION: What is the most comprehensive summary of PCA?

OPTIONS:

- ① PCA is a supervised learning method for classification
- ② PCA is an unsupervised technique that creates uncorrelated linear combinations of features, capturing variability, reducing dimensionality, and addressing multicollinearity
- ③ PCA is a clustering algorithm
- ④ PCA only works for categorical data

ANSWER: B

DETAILED EXPLANATION:

- **Unsupervised:** No labels needed
- **Linear combinations:** Creates new features
- **Uncorrelated:** PCs are orthogonal
- **Captures variability:** Ordered by variance explained
- **Reduces dimensionality:** Fewer features
- **Solves multicollinearity:** PCs are independent

WHY OTHER OPTIONS ARE WRONG:

- **A:** PCA is UNSUPERVISED
- **C:** PCA is for dimensionality reduction, not clustering
- **D:** PCA is for numerical data

EXAM TRAP: PCA = unsupervised dimensionality reduction with uncorrelated components!

QUESTION: What is the purpose of the training set?

OPTIONS:

- ❶ To select between competing models
- ❷ To evaluate final model performance
- ❸ To estimate model parameters (learning)
- ❹ To tune hyperparameters

ANSWER: C

DETAILED EXPLANATION:

- Used to estimate parameters (weights)
- The model "learns" from this data
- Largest portion of data typically
- Should represent the population

WHY OTHER OPTIONS ARE WRONG:

- **A:** Validation set is for model selection
- **B:** Test set is for final evaluation
- **D:** Validation set is for hyperparameter tuning

EXAM TRAP: Training set = **estimating parameters (learning)**!

QUESTION: What is the purpose of the validation set?

OPTIONS:

- ❶ To estimate model parameters
- ❷ To select between competing models and tune hyperparameters
- ❸ To test final model performance
- ❹ To increase data size

ANSWER: B

DETAILED EXPLANATION:

- Used for model selection
- Tune hyperparameters
- Compare competing models
- Once used, it's "contaminated"

WHY OTHER OPTIONS ARE WRONG:

- **A:** Training set estimates parameters
- **C:** Test set evaluates final performance
- **D:** Validation set doesn't increase data

EXAM TRAP: Validation set = model selection and hyperparameter tuning!

QUESTION: What is the purpose of the test set?

OPTIONS:

- ❶ To estimate model parameters
- ❷ To select between competing models
- ❸ To evaluate the final model's performance on unseen data
- ❹ To tune hyperparameters

ANSWER: C

DETAILED EXPLANATION:

- Held back until final evaluation
- Assess generalization performance
- Truly unseen data
- Provides unbiased performance estimate

WHY OTHER OPTIONS ARE WRONG:

- **A:** Training set estimates parameters
- **B:** Validation set selects models
- **D:** Validation set tunes hyperparameters

EXAM TRAP: Test set = final unbiased evaluation!

QUESTION: What is overfitting in ML?

OPTIONS:

- ❶ Model fits training data well but fails to generalize
- ❷ Model fits test data better than training data
- ❸ Model is too simple to capture patterns
- ❹ Model has no parameters

ANSWER: A

DETAILED EXPLANATION:

- Model captures noise, not signal
- Excellent training performance
- Poor test performance
- Too complex model

WHY OTHER OPTIONS ARE WRONG:

- **B:** This would be unusual and suspicious
- **C:** This is underfitting
- **D:** Overfitting involves too many parameters

EXAM TRAP: Overfitting = good training, poor test!

QUESTION: What is underfitting in ML?

OPTIONS:

- ❶ Model fits training data well but fails to generalize
- ❷ Model is too simple to capture patterns
- ❸ Model fits test data perfectly
- ❹ Model has too many parameters

ANSWER: B

DETAILED EXPLANATION:

- Too simple model
- Cannot capture relationships
- Poor performance on both training and test
- Opposite of overfitting

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is overfitting
- **C:** This would be suspicious
- **D:** This describes overfitting

EXAM TRAP: Underfitting = model too simple!

QUESTION: What is a common rule of thumb for sample splitting?

OPTIONS:

- ❶ Training: 50%, Validation: 25%, Test: 25%
- ❷ Training: 66%, Validation: 17%, Test: 17%
- ❸ Training: 80%, Validation: 10%, Test: 10%
- ❹ Training: 90%, Validation: 5%, Test: 5%

ANSWER: B

DETAILED EXPLANATION:

- Roughly two-thirds training
- Remaining third split equally between validation and test
- Approximate guideline
- May vary depending on data size

WHY OTHER OPTIONS ARE WRONG:

- **A:** Training is too small
- **C:** Validation/test too small
- **D:** Validation/test too small

EXAM TRAP: Common split = 66% training, 17% validation, 17% test!

QUESTION: How should cross-sectional data be split?

OPTIONS:

- ❶ Chronologically (first part training, last part test)
- ❷ Randomly from the total dataset
- ❸ Always use all data for training
- ❹ Sort by feature values

ANSWER: B

DETAILED EXPLANATION:

- Cross-sectional data has no order
- Random sampling ensures representativeness
- Avoids bias
- Each observation independent

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is for time-series data
- **C:** Need to hold out data for testing
- **D:** Should be random, not sorted

EXAM TRAP: Cross-sectional = **random split!**

QUESTION: How should time-series data be split?

OPTIONS:

- ❶ Randomly from the total dataset
- ❷ Chronologically: first part training, last part test
- ❸ Always use all data for training
- ❹ Sort by feature values

ANSWER: B

DETAILED EXPLANATION:

- Time-series has chronological order
- Training = earlier periods
- Validation = middle periods
- Test = most recent periods
- Avoids look-ahead bias

WHY OTHER OPTIONS ARE WRONG:

- **A:** Random split would break chronology
- **C:** Need to hold out data for testing
- **D:** Chronology matters, not feature values

EXAM TRAP: Time-series = **chronological split!**

QUESTION: When is cross-validation used?

OPTIONS:

- ❶ When there is abundant data
- ❷ When there is limited data
- ❸ Always
- ❹ Never

ANSWER: B

DETAILED EXPLANATION:

- Used when data is limited
- Combines training and validation
- Test data still held back
- More efficient use of data

WHY OTHER OPTIONS ARE WRONG:

- **A:** With abundant data, simple split works
- **C:** Only needed for limited data
- **D:** Cross-validation is often used

EXAM TRAP: Cross-validation = for limited data!

QUESTION: How does k-fold cross-validation work?

OPTIONS:

- ❶ Use all data for training once
- ❷ Split data into k folds, train on k-1, validate on 1, repeat k times
- ❸ Split data once into training and test
- ❹ Use all data for validation

ANSWER: B

DETAILED EXPLANATION:

- Combines training and validation
- k folds (e.g., 5 or 10)
- Each fold used as validation once
- Results averaged across folds

WHY OTHER OPTIONS ARE WRONG:

- **A:** This is just training
- **C:** This is simple split
- **D:** This is no training

EXAM TRAP: k-fold CV = train on k-1, validate on 1, repeat k times!

QUESTION: What indicates a model generalizes well?

OPTIONS:

- ❶ Training accuracy is much higher than test accuracy
- ❷ Test accuracy is similar to training accuracy
- ❸ Training accuracy is very low
- ❹ Test accuracy is very low

ANSWER: B

DETAILED EXPLANATION:

- Similar performance on training and test
- Model captures signal, not noise
- Not overfitting or underfitting
- Good predictive performance

WHY OTHER OPTIONS ARE WRONG:

- **A:** This indicates overfitting
- **C:** This indicates underfitting
- **D:** This indicates poor performance

EXAM TRAP: Good generalization = similar training and test performance!

QUESTION: What is the correct order of model development?

OPTIONS:

- ❶ Train → Test → Validate
- ❷ Train → Validate → Test
- ❸ Validate → Train → Test
- ❹ Test → Train → Validate

ANSWER: B

DETAILED EXPLANATION:

- ❶ Train: Estimate parameters
- ❷ Validate: Select model and tune hyperparameters
- ❸ Test: Evaluate final model performance
- ❹ Test data used only once at the end

WHY OTHER OPTIONS ARE WRONG:

- ❶ **A:** Test should be last
- ❸ **C:** Training must come before validation
- ❹ **D:** All steps are out of order

EXAM TRAP: Correct order = Train → Validate → Test!

QUESTION: Which programming languages are popular for ML?

OPTIONS:

- ❶ R and Python
- ❷ Java and C++
- ❸ HTML and CSS
- ❹ SQL and PHP

ANSWER: A

DETAILED EXPLANATION:

- **R:** Great for statistics and visualization
- **Python:** General-purpose, great for ML
- Both open-source, cross-platform
- Rich ecosystem of libraries

WHY OTHER OPTIONS ARE WRONG:

- **B:** Java/C++ are less common for ML
- **C:** HTML/CSS are web languages
- **D:** SQL/PHP are database/web languages

EXAM TRAP: Popular ML languages = R and Python!

QUESTION: Which Python library is for machine learning?

OPTIONS:

- ❶ NumPy
- ❷ Scikit-Learn
- ❸ Matplotlib
- ❹ All of the above

ANSWER: D

DETAILED EXPLANATION:

- **NumPy:** Array operations, linear algebra
- **Scikit-Learn:** ML models, preprocessing
- **Matplotlib:** Data visualization
- All part of Python ML ecosystem

WHY OTHER OPTIONS ARE WRONG:

- **A:** Correct but incomplete
- **B:** Correct but incomplete
- **C:** Correct but incomplete

EXAM TRAP: All are **important Python libraries** for ML!

QUESTION: What is the most comprehensive summary of ML fundamentals?

OPTIONS:

- ① ML is only about supervised learning
- ② ML includes supervised, unsupervised, semi-supervised, and reinforcement learning, with data pre-processing, PCA, and proper train/validation/test splits
- ③ ML only uses Python
- ④ ML doesn't require data cleaning

ANSWER: B

DETAILED EXPLANATION:

- **ML Types:** Supervised, Unsupervised, Semi-supervised, Reinforcement
- **Data Pre-processing:** Cleaning, scaling, encoding
- **PCA:** Dimensionality reduction
- **Splitting:** Training, Validation, Test

WHY OTHER OPTIONS ARE WRONG:

- **A:** ML includes more than supervised
- **C:** R is also used
- **D:** Data cleaning is essential

EXAM SUCCESS: Understand ML types, data pre-processing, PCA, and data splitting!